

Blockchain Hash Function

A hash function is a mathematical function that takes an input string of any length and converts it to a fixed-length output string. The fixed-length output is known as the hash value

Problem: Building a Decentralized Domain Name System (DNS)

The domain name system (DNS) is **a naming database in which internet domain names are located and translated into Internet Protocol (IP) addresses**. The domain name system maps the name people use to locate a website to the IP address that a computer uses to locate that website.

Solution: In a traditional DNS system, domain names are resolved to IP addresses through a centralized server. However, we can leverage the benefits of both Hash Tables and Blockchain to create a decentralized DNS system.

- Hash Table for Domain Name Lookup: We can use a Hash Table to store domain names and their corresponding IP addresses. Each node in the network maintains a local Hash Table, mapping domain names to IP addresses. When a user wants to access a website, they query their local Hash Table for the IP address associated with the domain name.
- Blockchain for Consensus and Immutable Records: To ensure the integrity and decentralization of the DNS system, we can use a Blockchain to reach consensus on the domain name records. Each time a node updates its Hash Table with a new domain name and IP address mapping, it creates a transaction containing the update and adds it to a block. The block is then appended to the blockchain.

- **Consensus Mechanism:** To achieve consensus on the DNS updates, we can utilize a suitable consensus mechanism such as Proof of Work (PoW) or Proof of Stake (PoS). Miners or validators in the network compete to add the next block to the blockchain by solving a computational puzzle or being chosen based on their stake. This ensures that updates to the DNS system are agreed

upon by the majority of participants.

- **Updating the Hash Tables:** When a new block is added to the blockchain, nodes in the network update their local Hash Tables to reflect the latest DNS mappings. By replicating the blockchain across multiple nodes, the DNS system becomes decentralized and resistant to single points of failure.

This combined approach of using a Hash Table for fast domain name lookup and a Blockchain for decentralized consensus ensures a reliable and tamper-proof DNS system. It provides the benefits of efficient data retrieval from the Hash Table while leveraging the security and decentralization properties of the Blockchain.